

ERC usage

LVS/RC section

Version 1.2

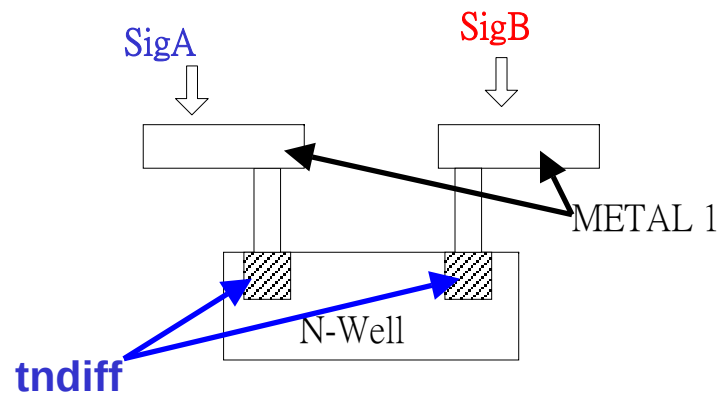
June 07, 2007

What is ERC (Electrical Rule Checking)

- ERC is a special option for designers. Some errors can be waived, but others may be deadly errors. So, designers must review every error or warning for LVS/ERC report.
- Some ERC rules are included in TSMC standard LVS command files.
 - Soft connect checking
 - Path checking
 - ptap/ntap checking
 - MOS s/d power&ground checking
 - Gate directly connects to power or ground.
 - Floating gate
 - Floating well

Soft connect and soft check

- N-well and P-well are high resistor material, so



1. Treat as Short

Sig A connect to Sig B

--> If Sig A is a power line and Sig B connect to a IP power.
--> The IP get a high resistor power, it means IR drop is very serious.

----> **ERROR**

2. Treat as open

Sig A does not connect to Sig B

--> If Sig A is a power signal and sig B is ground signal
--> power and ground short

----> **ERROR**

There are two methods to solve this problem

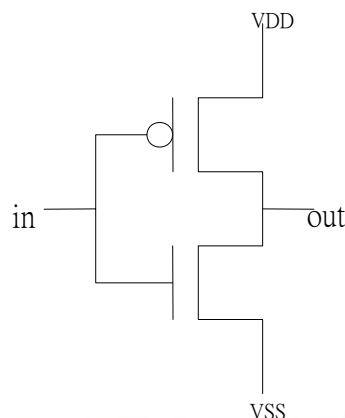
1. Run two LVS command file, one is “treat as short”, the other is “treat as open”.
2. Use soft connect and soft check!!!

What is soft connect and soft checking ?

- *Sconnect definition*: Passes established connectivity from the upper layer polygons onto the specified lower layer polygons (one-directional) < calibre manu.>
- *Softchk* : Help to search which contact connect to WELL.
- TSMC LVS/ERC command file contain those option, and those errors/warnings are report in lvs.rep or lvs.rep.ext .
- Please confirm every error or warning of those three file ~.rep ~.rep.ext svdb/~.rep .
- Give user a warning on the lvs.rep.ext file. User can debug this error by “calibre -rve” and open the svdb/~.softchk file .

Path Checking

- 4 kinds of path checking
 - 1. Nodes with a path to power but not ground
 - 2. Nodes with a path to ground but not power
 - 3. Nodes without a path to both path and groundout
 - 4. Nodes without a path to pin
- Purpose
 - Help designer to check if the circuit miss something.



Pathchk Report (svdb/~.Rep)

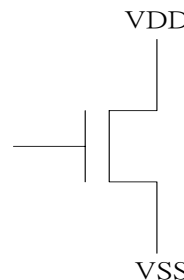
```
File Edit Tools Syntax Buffers Window
ERC PATHCHK REPORT for ERC_check
PATHCHK GROUND && ! POWER:
  11
PATHCHK POWER && ! GROUND:
  8
PATHCHK ! POWER && ! GROUND NOFLOAT:
  13, 14, 19, 23, 24, 29
PATHCHK ! LABELLED NOFLOAT:
  19, 23, 24, 29
2
2
2
2
2
```

ERC Report

- Get ERC report in LVS stage.
- Calibre : %calibre -lvs -spi layout.net lvs_command_file
 - Calibre please check the “calibre_erc.db” and “calibre_erc.sum” files.
- Hercules : %hercules lvs_command_file
 - Hercules please check the “Top.LAYOUT_ERRORS” file.
- Assura : %assura LVS.rsf
 - Assura please check the “Top.err” file.

PTAP/NTAP and Special MOS Connectivity Checking

- **PTAP connect to power/ NTAP connect to ground**
 - Check ERC errors “PPVDD49” for PTAP
 - Check ERC errors “NPVSS49” for NTAP
- **N/P MOS source/drain one connect to POWER the other connect to GROUND**
 - Check ERC errors “mppg” for PMOS.
 - Check ERC errors “mnpng” for NMOS.
- Calibre please check the “calibre_erc.db” and “calibre_erc.sum” files.
- Hercules please check the “Top.LAYOUT_ERRORS” file.
- Assura please check the “Top.err” file.



Gate directly connects to power or ground

- **PMOS gate directly connects to power without the 2.5v and 3.3v IO devices.**
 - For ESD protection, if core or 1.8v PMOS gate directly connect power node, gate will be damaged.
 - Check ERC errors “ppvdd150” for PMOS gate.
- **NMOS gate directly connects to ground without the 2.5v and 3.3v IO devices.**
 - For ESD protection, if core or 1.8v NMOS gate directly connects ground node, gate will be damaged.
 - Check ERC errors “npvss150” for NMOS gate.
- **Calibre please check the “calibre_erc.db” and “calibre_erc.sum” files.**
- **Hercules please check the “Top.LAYOUT_ERRORS” file.**
- **Assura please check the “Top.err” file.**

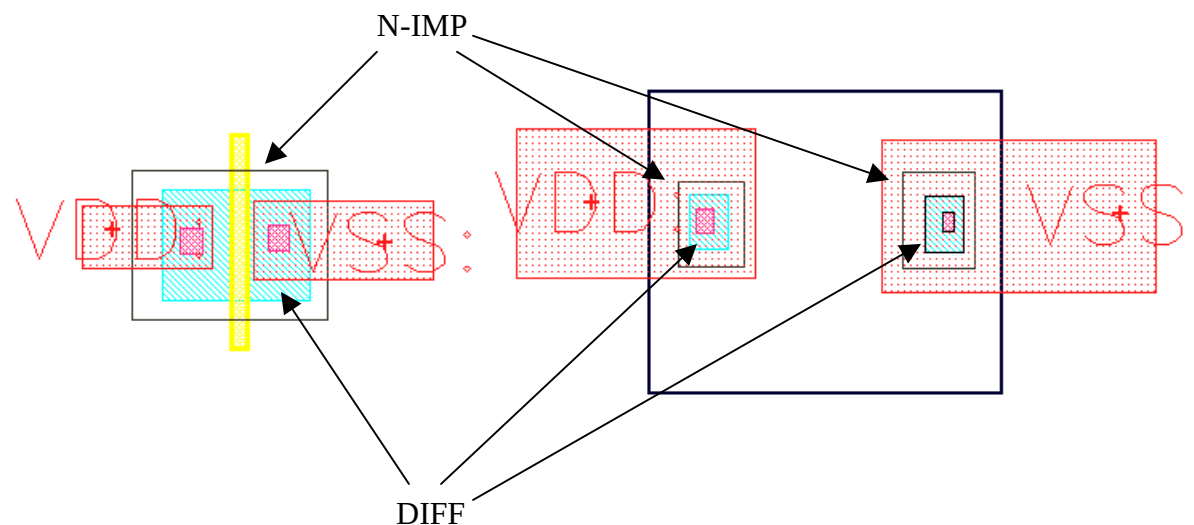
Floating gate

- There is no any Contact interact with the poly gate.
 - Check ERC errors “floating” for all gate.
- Calibre please check the “calibre_erc.db” and “calibre_erc.sum” files.
- Hercules please check the “Top.LAYOUT_ERRORS” file.
- Assura please check the “Top.err” file.

Floating well

- There is no any power or ground connect to NWEL or PSUB.
 - Check ERC errors “floating.nxwell” for NWEL.
 - Check ERC errors “floating.psub” for PSUB.
- Calibre please check the “calibre_erc.db” and “calibre_erc.sum” files.
- Hercules please check the “Top.LAYOUT_ERRORS” file.
- Assura please check the “Top.err” file.

A Layout Example



~.sum

--- RULECHECK RESULTS STATISTICS

RULECHECK mppg TOTAL Result Count = 1
RULECHECK mnpd TOTAL Result Count = 0
RULECHECK ppvdd49 ... TOTAL Result Count = 0
RULECHECK npvss49 ... TOTAL Result Count = 1

~.ext

WARNING: Stamping conflict in SCONNECT -
Multiple source nets stamp one target net.
Use LVS REPORT OPTION S or LVS
SOFTCHK statement to obtain detailed information.

Hercules ERC Check Error File

- Please check the Top.LAYOUT_ERRORS file.

```
File Edit Tools Syntax Buffers Window
Structure name: ERC_check

ERROR SUMMARY

No Comment
ERR_TEXT_SHORT_DISCARD ..... 2 violations found.

No Comment
C_THRU nplug INSIDE nxwell { } (1;0) ..... 2 violations found.

No Comment
C_THRU pplug INSIDE psub_term { } (1;1) ..... 2 violations found.

No Comment
C_THRU pplug INSIDE psub { } (1;2) ..... 2 violations found.

No Comment
C_THRU n_pplug INSIDE n_psub { } (1;4) ..... 2 violations found.

No Comment
C_THRU dnwc INSIDE DNW { } (1;6) ..... 2 violations found.

No Comment
ERR_TEXT_SHORT_DISCARD ..... 2 violations found.

No Comment
ERR_TEXT_SHORT_DISCARD ..... 2 violations found.

No Comment
NET_PATH_CHECK { } PERM=path1_Out(1;13) ..... 13 violations found.

No Comment
NET_PATH_CHECK { } PERM=path3_Out(1;15) ..... 16 violations found.

No Comment
NET_PATH_CHECK { } PERM=path4_Out(1;16) ..... 16 violations found.
```

Assura ERC Check Error File

- Please check the Top.err file.
- Currently the Assura can not support the net-path check function.

```
File Edit Tools Syntax Buffers Window
=====
Rule No. 2 : nxwell_StampErrorMult
Real Error Count : 1; Flat Error Count : 1
=====

Rule No. 3 : psub_StampErrorMult
Real Error Count : 1; Flat Error Count : 1
=====

Rule No. 7 : DNW_StampErrorMult
Real Error Count : 1; Flat Error Count : 1
=====
2
2
2
2
2
```

Summary for ERC checker

- Please confirm every error or warning of these three files --- “~.rep” “~.rep.ext” “svdb/~.rep” for Calibre.
- Please confirm the Top.LAYOUT_ERRORS file for Hercules.
- Please confirm the Top.err file for Assura.
- Every soft connect error must be fixed.
- Other ERC errors/warnings need to be reviewed by circuit designer.